

Active Learning for Learning to Defer

HCAI Project 3

AG News Classification and User Study

Abstract

In this project, I worked on a classification system where the AI model does not always have to make the final decision by itself. The main idea was to explore situations where the model can ask for expert help when it is not confident enough. I used the AG News dataset and built a baseline classifier using TF IDF and Logistic Regression. After that, I added a simulated expert, a confidence based Learning to Defer approach and an active learning experiment. In the final part, I designed a Streamlit based user study where participants can see an AI recommendation, decide whether to trust it and ask for expert help if they want. The aim of the project was to combine the technical machine learning part with a more human centred interaction design.

1. Introduction

Usually, when we train a machine learning model, we mainly focus on making the prediction as accurate as possible. During this project, I wanted to look at another question. What should the model do when it is not sure about its prediction? Instead of forcing the model to answer every time, it can be useful to involve an expert for difficult cases.

This is the main idea behind Learning to Defer. The model first looks at an input and its confidence. If the confidence is high enough, it makes the prediction. If the case is uncertain, the prediction can be passed to an expert. I found this interesting because it represents a more realistic human AI collaboration scenario.

2. Dataset and Baseline Model

For the project, I used the AG News dataset. The task is to classify news articles into four categories: World, Sports, Business and Sci Tech. I used TF IDF to convert the text into numerical features and Logistic Regression as the baseline classifier.

The baseline model also provides prediction probabilities. These probabilities are important later because I use them as a simple measure of how confident the model is about a prediction. The baseline gives a starting point for comparing the Learning to Defer approach.

3. Simulated Expert

The next part of the project was to add an expert. Since a real expert was not available for this project, I simulated expert labels. The expert was not designed to be perfect. This was important because otherwise there would be no real decision about when to use the expert. In the simulation,

the expert performs better for some categories and can make mistakes for more confusing cases.

The purpose of this part was to create a situation where both the classifier and the expert have different strengths. This makes the later deferral decision more meaningful.

4. Learning to Defer

For Learning to Defer, I used the confidence of the classifier. If the confidence is below a selected threshold, the example can be sent to the expert. If the confidence is above the threshold, the classifier keeps its own prediction.

I also considered a cost for asking the expert. This is useful because, in a real situation, expert time is limited. If there was no cost, the system could simply send every difficult example to the expert. The threshold therefore creates a balance between using the model independently and asking for help.

I evaluated the approach using classifier accuracy, combined accuracy and the deferral rate. I was especially interested in checking whether the deferred examples were actually difficult examples where the classifier was more likely to make mistakes.

5. Active Learning

In the active learning part, I compared uncertainty sampling with random sampling. With uncertainty sampling, the system selects examples where the classifier is less confident. With random sampling, examples are selected randomly.

The idea was to check whether expert queries can be used more efficiently. My expectation was that uncertainty sampling should be more useful when the query budget is limited because it focuses on examples that are likely to provide more information.

6. User Study Design

Initially, the Streamlit interface mainly showed technical information such as baseline accuracy, expert performance, confidence thresholds and active learning results. After considering the project requirement, I separated the participant interface from the technical implementation.

The project now has a landing page and a separate user study. The landing page introduces the project, provides the report and contains a button to start the study. This keeps the participant flow simple.

Before starting the study, participants read a short explanation and provide consent. They then complete ten classification trials. For every trial, the participant reads a news article and sees the AI recommendation together with its confidence.

7. Participant Interaction

The participant has three main choices. They can accept the AI prediction, ask the expert for help, or reject the AI prediction and select another category. When expert help is requested, the expert recommendation becomes visible and the participant can make the final choice.

After making the final decision, the participant also rates how confident they are and how difficult they found the article. These questions were included because I wanted to capture not only the final answer but also how the participant experienced the interaction.

The true label is not shown during the study. This is important because showing the correct answer could influence the participant's later decisions.

8. What Can Be Analysed

The study records the AI prediction, AI confidence, participant decision, whether expert help was requested, final prediction, participant confidence and perceived difficulty. Based on this information, different behaviours can be analysed.

For example, it is possible to check whether participants trust the AI more when its confidence is high. It is also possible to see when participants ask for expert help and whether difficult articles lead to more deferral decisions. Another interesting comparison is between the AI recommendation and the participant's final decision.

9. Limitations

There are some limitations in the current project. The expert is simulated, so it cannot fully represent a real human expert. The user study contains a small number of trials and is mainly designed as a reproducible prototype. Also, participant responses are currently stored for the session and can be downloaded as a CSV file.

Even with these limitations, the project provides a complete example of how a technical Learning to Defer pipeline can be connected with a participant facing interface.

10. Conclusion

This project helped me explore machine learning from both a technical and human centred perspective. I started with a baseline text classifier and then added a simulated expert, Learning to Defer and active learning. In the final stage, I created a separate user study where participants can interact with the AI system.

The main thing I learned from this project is that a good AI system does not always need to make every decision independently. It can also be useful for a system to identify difficult cases and involve another source of expertise. The final interface and report bring these different parts

together into one complete project.